

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2023****Course: B. Tech. Branch: ENT/Communication Engineering Semester: V****Subject Code & Name: [BTETC501 /BTEXC501] Electromagnetic Field Theory****Max Marks: 60****Date: 07/08/2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level)	Marks
Q. 1 Solve Any Two of the following.		12
A) Given that vector $\vec{rA} = -4\vec{ax} - 2\vec{ay} + \vec{az}$ $\vec{rB} = 8\vec{ax} + 2\vec{ay} + 5\vec{az}$ and at point C (1,3,4) Find (i) $\vec{rAB}$ (ii) $\vec{rAC}$ (iii) unit vector directed from point C to A	Understand	6
B) Given two points C (-3,2,1) and D (5, 20°, -70°). Find (i) Spherical co-ordinates of C (ii) Cartesian co-ordinates of D (iii) the distance from point C to D	Understand	6
C) State and explain the Coulomb's law and the superposition principle applied for the static charge distribution.	Understand, Remember	6
Q.2 Solve Any Two of the following.		12
A) State Biot-Savart Law and explain its use in determining the magnetic field around a current-carrying conductor.	Understand, Remember	6
B) State the continuity equation for electric current and its significance in electromagnetism.	Understand, Remember	6
C) Consider a magnetic field propagating through two different materials: material A and material B. Material A has a magnetic permeability of $\mu_A = 4\pi \times 10^{-7}$ H/m, and material B has a magnetic permeability of $\mu_B = 2\pi \times 10^{-7}$ H/m. The magnetic field intensity (H) in material A is given as $H = 150$ A/m, and the magnetic field is incident on the interface between material A and material B at an angle of 60 degrees with the normal to the interface. Find the reflected magnetic field intensity (H_r) and the transmitted magnetic field intensity (H_t) at the interface.	Understand	6
Q. 3 Solve Any Two of the following.		12
A) State and explain the basic Maxwell's Equations in its point and integral form with their proper significance.	Understand	6
B) Define and explain the boundary conditions for electromagnetic fields at the interface of two different materials.	Understand	6
C) Describe the mathematical representation of a time harmonic electromagnetic field in both the time domain and phasor domain. Also discuss relationship between them.	Understand	6

Q.4 Solve Any Two of the following.		12
A) A transmission line cable has following primary constants. $R = 75 \Omega/\text{km}$, $G = 60 \mu\text{mho}/\text{km}$, $L = 1.7 \text{ mH}/\text{km}$ and $C = 0.075 \mu\text{F}/\text{km}$, at a signal of 2kHz, calculate (i) characteristics impedance, (ii) attenuation constant (iii) phase constant (iv) wavelength (v) velocity and (vi) time taken by signal to travel 100 km along line.	Understand	6
B) What is smith chart? Write the procedure to find the impedance from the given admittance using smith chart and List the applications of smith chart	Understand, Remember	6
C) Explain some applications of Transmission lines	Understand	6
Q. 5 Solve Any Two of the following.		12
A) State the Poynting theorem and vector and describe how it represents the flow of electromagnetic energy in space. Discuss the physical interpretation of the Poynting vector.	Understand, Remember	6
B) Explain the law of reflection for plane waves. Write down the mathematical expression that represents the law of reflection.	Understand	6
C) Explain Linear, circular & Elliptical polarization in details. A plane wave is traveling in a medium with a refractive index of 1.6 and is incident on a boundary with another medium with a refractive index of 1.2. If the angle of incidence is 50 degrees, calculate the angle of reflection and the angle of refraction	Understand	6

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